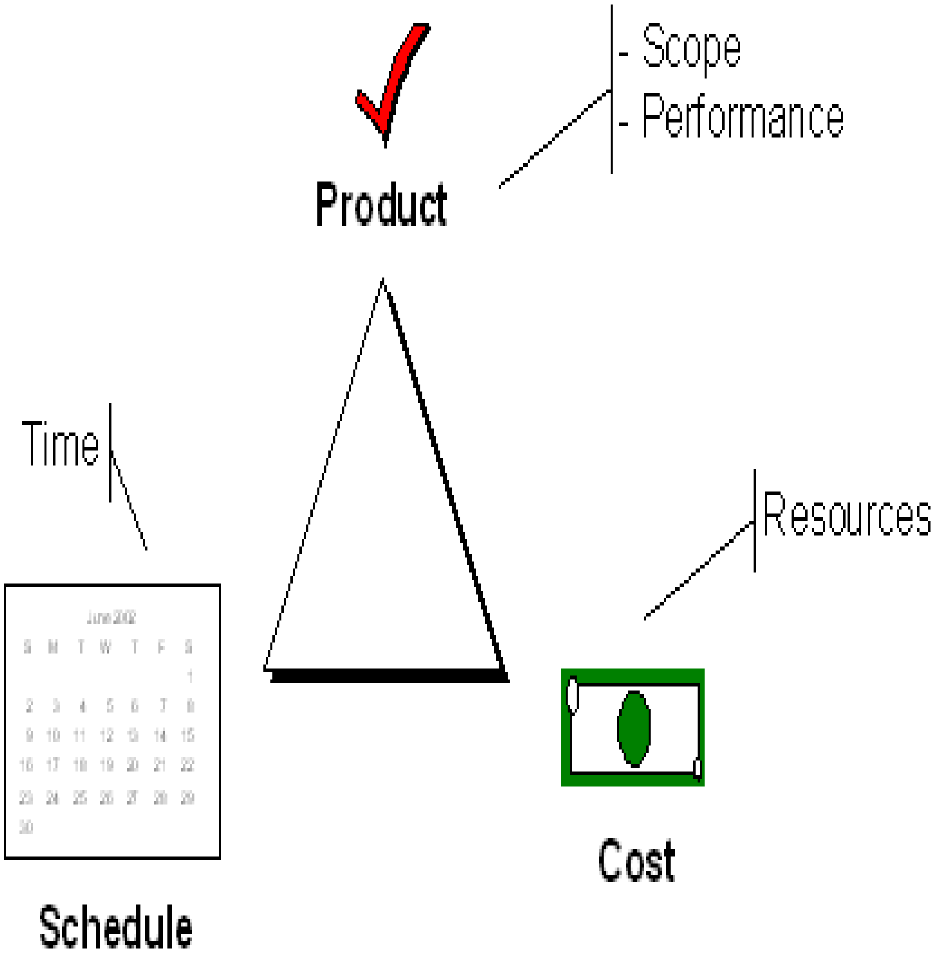
Trade-off Triangle

- Fast, cheap, good. Choose two.

- Know which of these are fixed & variable for every project



PM Value Calculator

of people on the teams (avg)

x

hours saved / person / day

x

5 days / week

x

# of weeks in the project

x

INR / hour (salary + benefits)

= INR / project

x

projects per year

=

Annual Potential Productivity Opportunity

The Big Picture

Total hours saved = 1 hr /prs /day x 5 days /wk x 52 wks x10 prs x 11 prjs

Project Management Value Calculator					
# of projects		11	Avg. Hourly rate (INR)		100
Hrs saved / person / day		1	Days / week		5
	People				
Duration	5	10	15	20	25
3 months	357500	715000	1072500	1430000	1787500
6 months	715000	1430000	2145000	2860000	3575000
9 months	1072500	2145000	3217500	4290000	5362500
12 months	1430000	2860000	4290000	5720000	7150000
15 months	1787500	3575000	5362500	7150000	8937500
18 months	2145000	4290000	6435000	8580000	10725000
21 months	2502500	5005000	7507500	10010000	12512500
24 months	2860000	5720000	8580000	11440000	14300000
30 months	3575000	7150000	10725000	14300000	17875000
36 months	4290000	8580000	12870000	17160000	21450000

The above computation for **small to medium size software companies**

Schedule

- List of tasks
  - With dates
  - With assigned resources (people)
  - With durations (expected time)
  - With predecessors and successors
- How do you get buy-in (e.g., commitment) from the team for a schedule?
  - History - Use data from past projs.
  - Increments – track progress

Scheduling Rules of Thumb

- One person - Schedule edit rule
  - If you have two people that need to, create a mechanism to link them together.
- Keep it simple and useful
- Level your resources
- Share schedule with your team
- 40-20-40; coding is 20% of the effort

Scheduling Steps

- Add in all the tasks (preferably in a hierarchy)
- Add in all the dependencies
- Break down large tasks into smaller tasks.
  - Optimally (one possible opinion) you want to schedule so the duration of each smallest task is at most 3-5 days
- Assign people (resources) to tasks
- Level your resources

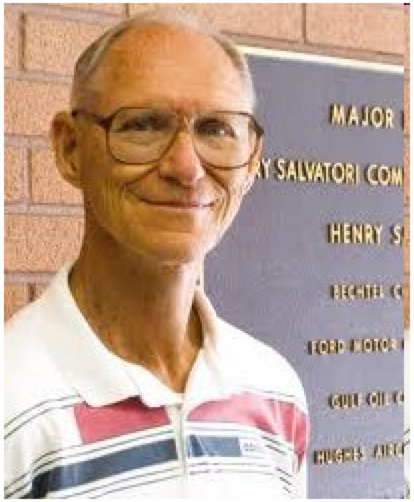
Resource Leveling

- A process to examine a project for an unbalanced use of people and to resolve over-allocations or conflicts
- Happens when multiple tasks are scheduled at the same time for the same person
- Solution: Introduce "**fake**" dependencies, **Split resource usage**

Understanding the Problem

At the beginning you should ask yourself these questions

- **Why** is the system being developed?
- **What** will be done?
- **When** will it be accomplished?
- **Who** is responsible?
- **Where** are they organizationally located?
- **How** will the job be done **technically** and **managerially**?
- How much of **each resource** (e.g., people, software, tools, database – **Which**?) will be **needed**?



Barry Boehm

Define success and failure

- Don’t lie to yourself! – An ethical issue, !SE.
- Be confident, trust yourself for success!
- **Quantify your project outcomes** to allow success or failure
- A vague or un-measurable outcome is much less helpful

Scheduling

- One of the most important things you can do is **schedule**.
- Also **one of the first things** you should do!
- Many Tools exist – We will look into one of it.

Planning

- **Time is the key resource** – requires carefully managing it.
- You must **begin planning immediately**
  - Given limited information - Plan anyway and then revise.
- Major step – **Creating a plan**.
  - **Software project scope** must be **unambiguous and understandable** at the management and technical levels.
  - Problem Decomposition: Sometimes called *partitioning* or *problem elaboration*

- PERT/CPM Charts
- Time-scale Charts

Project Evaluation and Review Technique (PERT)

What it is?
When to use it?

- A PERT chart is a graphical representation of a project's schedule.
  - Used as a project management tool.
  - Used in planning and tracking software projects.
- It shows sequence of tasks, which tasks can be performed simultaneously, and critical path of tasks that must be completed on time in order for project to meet its completion deadline.
- The chart can be constructed with a variety of attributes, such as
  - earliest and latest start times for each task,
  - earliest and latest finish times for each task, and
  - slack time between tasks.
- It can document an entire project or a key phase of a project.
- **Benefits:**
  - The chart allows a team to avoid unrealistic timetables and schedule expectations,
  - To help identify and shorten tasks that are bottlenecks, and to focus attention on most critical tasks.

Process to construct PERT charts

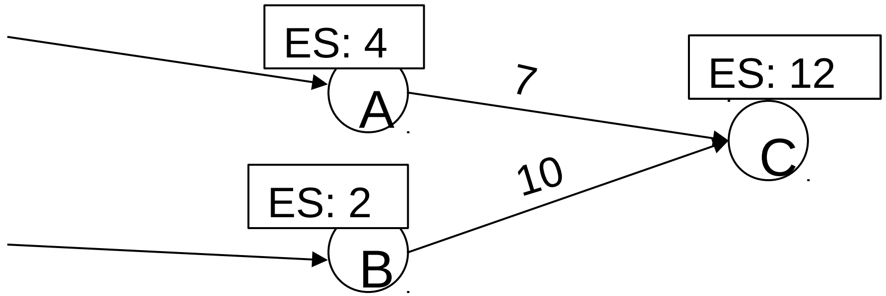
- Identify all tasks or project components.
  - Make sure the team includes people with first hand knowledge of the project so that during the brainstorming session all component tasks needed to complete the project are captured.
  - Document the list of tasks.
- Identify the first task that must be completed.
  - Place the appropriate task at the extreme left of the chart.
- Identify other tasks that can be started along with task #1.
  - Align these tasks either above or below task #1 on the chart.
- Identify the next task that must be completed.
  - Select a task that must wait to begin until task #1(or a task that starts simultaneously with task #1) is completed.
  - Place the appropriate next task to the right of the preceding task.
- Identify other tasks that can be started along with task #2.
  - Align these tasks either above or below task #2 on the chart.
- Continue this process until all component tasks are sequenced.

Process to construct PERT charts …cont’d

- Identify task durations.
  - Using the knowledge of team members, reach a consensus on the most likely amount of time each task will require for completion.
  - Document this time for each task.
- Use this information to construct the PERT chart.
  - Number each task, draw connecting arrows, and add task characteristics such as duration, anticipated start time, and anticipated end time.
- Determine the critical path.
  - The project’s critical path includes those tasks that must be started or completed on time to avoid delays to the total project.
  - Critical paths are typically displayed in red.

Computing ES → Critical Path

- Draw a network diagram of activities
- ES - earliest time the activity can start
- Determine the Early Start (ES) of each node.
  - Work from beginning node (ES=0) to final node
- **ESNode = Max(ESPrevNode + DurationPrevNode)**



Compute ES(C)

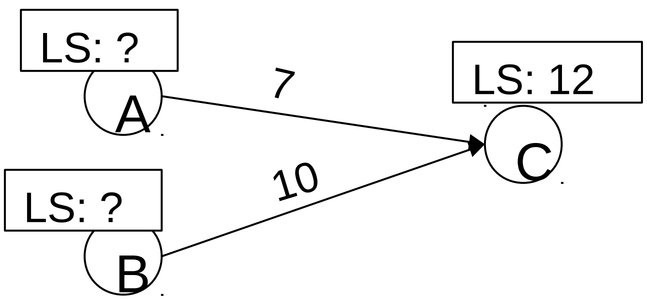
ES(A)+dur(A) = 4 + 7 = 11

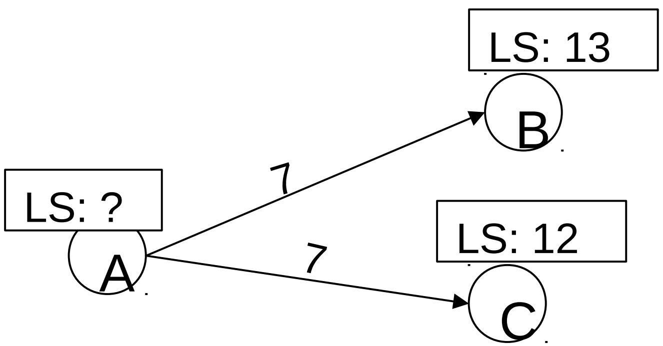
ES(B)+dur(B) = 2 + 10 = 12

Maximum (11,12) = 12 = ES(C)

Computing LS → Critical Path

- Determine the Late Start (LS) of each node.
  - Work from the final node to the beginning node.
  - The latest time the activity can start without changing the end date of the project
  - **LSNode = MIN(LSNextNode - DurationPrevNode)**
  - **Remember:** For the last node LS = ES





Compute LS(A), LS(B)
LS(C) - dur(A) = 12 - 7 = 5 = LS(A)
LS(C) - dur(B) = 12 - 10 = 2 = LS(B)

Compute LS(A)
LS(B) - dur(A) = 13 - 7 = 6
LS(C) - dur(A) = 12 - 7 = 5
Min(6,5) = 5 = LS(A)

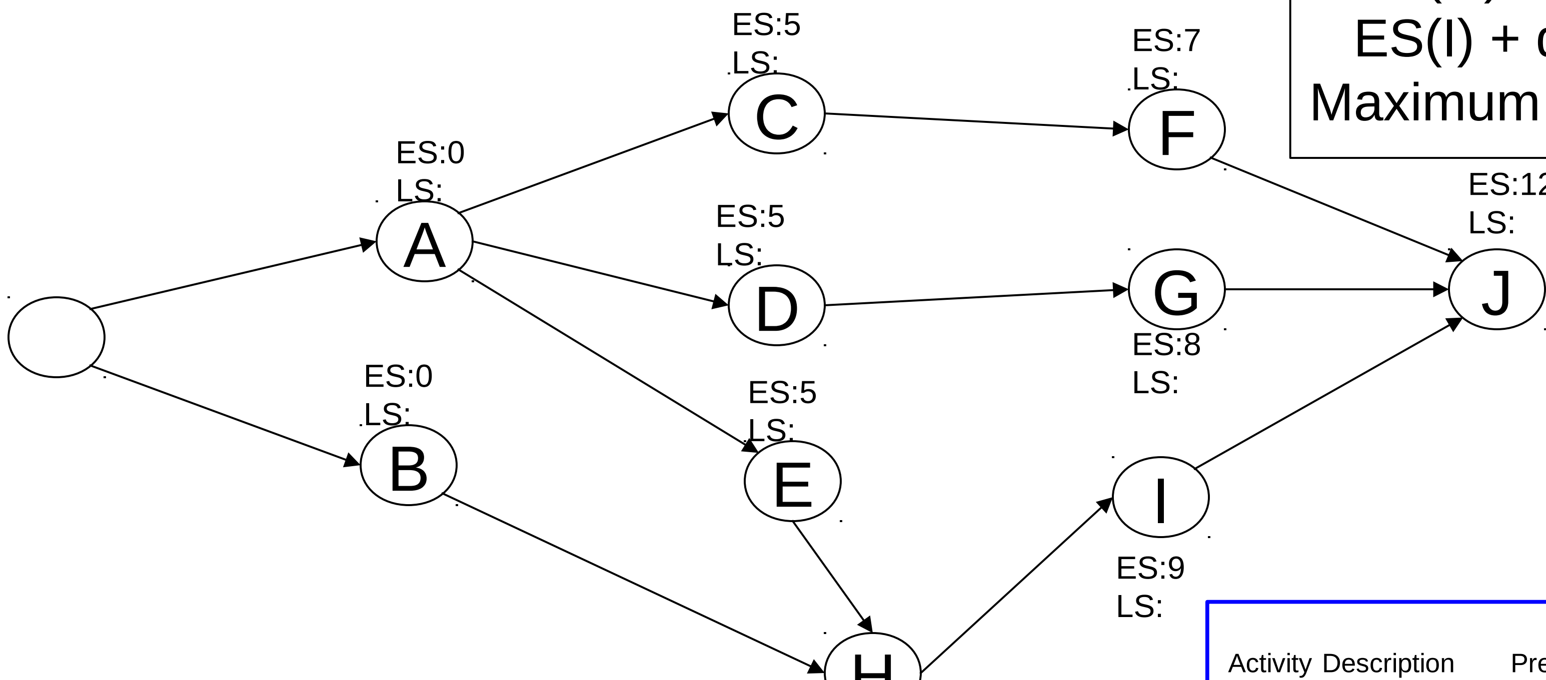
Example Project Specification

Activity	Description	Predecessor	Duration
A	Product design	(None)	5 months
B	Market research	(None)	1
C	Production analysis	A	2
D	Product model	A	3
E	Sales brochure	A	2
F	Cost analysis	C	3
G	Product testing	D	4
H	Sales training	B, E	2
I	Pricing	H	1
J	Project report	F, G, I	1

Example Node Network

Compute ES(J)
- ES(F) + dur(F) = ?
- ES(G) + dur(G) = ?
- ES(I) + dur(I) = ?

Maximum = ? = ES(J)



Compute ES(H)
- ES(E)+dur(E) = 5 + 2 = 7
- ES(B)+dur(B) = 0 + 1 = 1

Maximum = 7 = ES(H)

Activity	Description	Predecessor	Duration
A	Product design	(None)	5 months
B	Market research	(None)	1
C	Production	A	2
D	Product model	A	3
E	Sales brochure	A	2
F	Cost analysis	C	3
G	Product testing	D	4
H	Sales training	B, E	2
I	Pricing	H	1
J	Project report	F, G, I	1

Example Node Network

Compute LS(A)

LS(C) – dur(A) = 7 – 5 = 2

LS(D) – dur(A) = 5 – 5 = 0

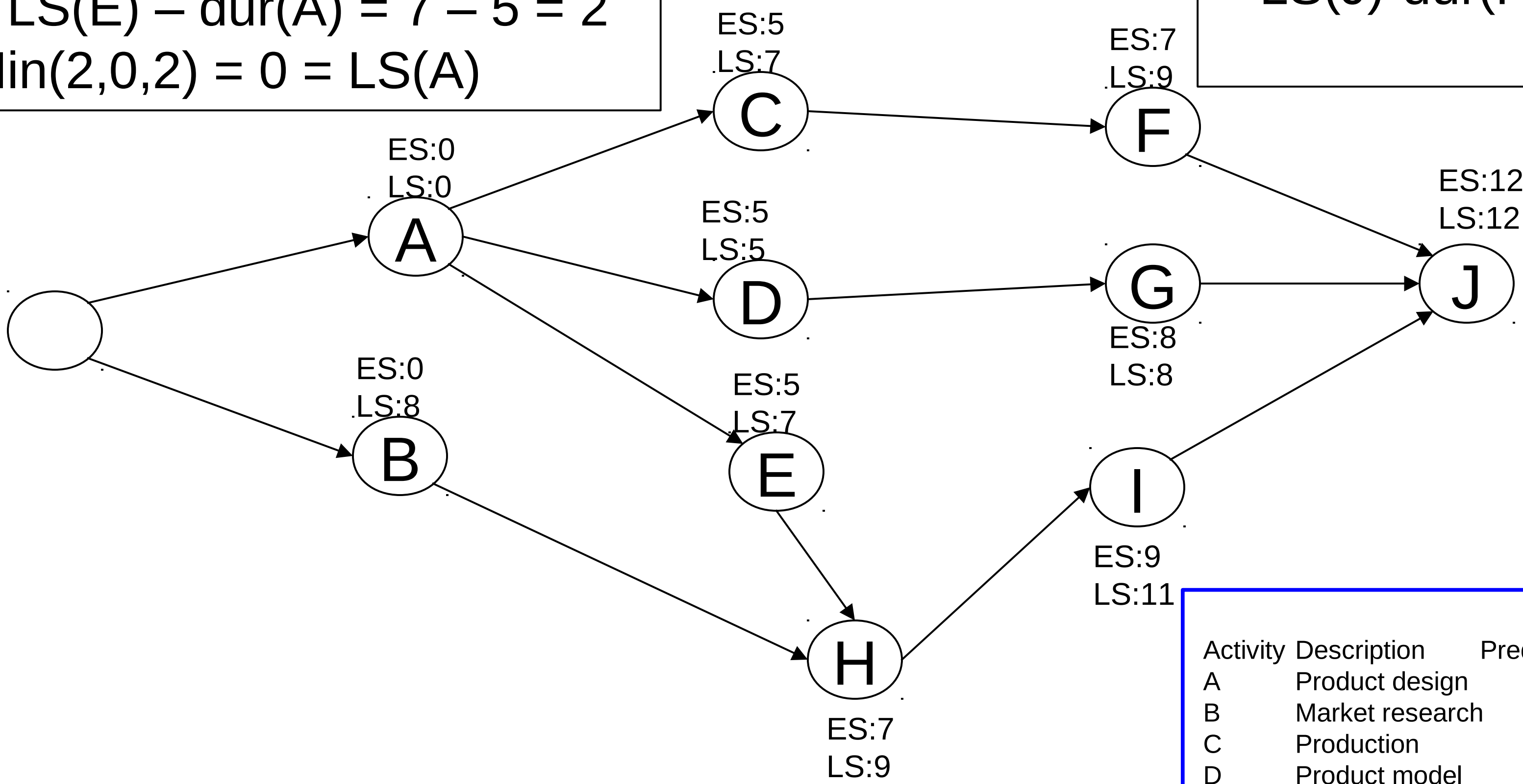
LS(E) – dur(A) = 7 – 5 = 2

Min(2,0,2) = 0 = LS(A)

Compute LS(F)

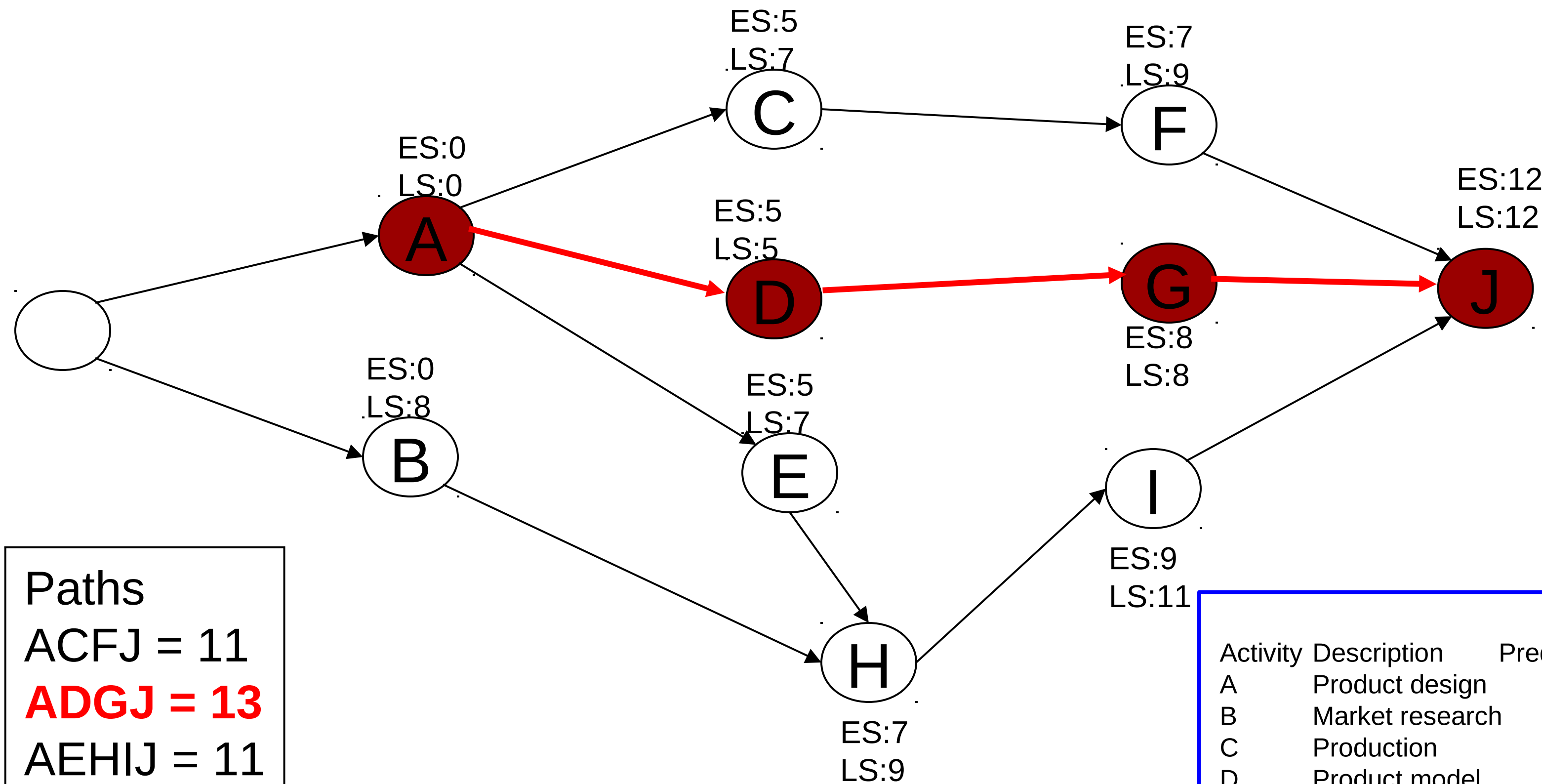
LS(J)-dur(F) = 12 – 3

= 9



Activity	Description	Predecessor	Duration
A	Product design	(None)	5 months
B	Market research	(None)	1
C	Production	A	2
D	Product model	A	3
E	Sales brochure	A	2
F	Cost analysis	C	3
G	Product testing	D	4
H	Sales training	B, E	2
I	Pricing	H	1
J	Project report	F, G, I	1

Example Node Network



Paths
ACFJ = 11
ADGJ = 13
AEHIJ = 11
BHIJ = 5

The Critical Path

Activity	Description	Predecessor	Duration
A	Product design	(None)	5 months
B	Market research	(None)	1
C	Production	A	2
D	Product model	A	3
E	Sales brochure	A	2
F	Cost analysis	C	3
G	Product testing	D	4
H	Sales training	B, E	2
I	Pricing	H	1
J	Project report	F, G, I	1